

MSPA – 3

Timing Analysis and Constraint-Based Optimization

Course: Synthesis and Optimization of VLSI Circuits

Assignment Title: Static Timing Analysis and Constraint-Based Optimization using Verilog

Tools: Vivado Design Suite, Verilog HDL

Objective

The objective of this assignment is to understand **Static Timing Analysis (STA)** and how **timing constraints affect synthesized hardware performance**.

Students will analyze the effect of clock constraints, input/output delays, and optimization directives on:

- Critical path delay
- Slack
- Maximum operating frequency
- Hardware resource utilization

Instructions

- Write **synthesizable Verilog HDL only**.
- Do **not use delays (#)** or non-synthesizable constructs.
- Create a **testbench for functional verification**.
- Perform **RTL synthesis and implementation**.
- Apply **timing constraints using XDC file**.
- Generate **timing and utilization reports**.
- Record observations and attach **waveform screenshots**.

Problem 1: Baseline Design and Timing Analysis

Task

Design a **4-bit Arithmetic Logic Unit (ALU)** that supports:

- Addition
- Subtraction
- AND
- OR

Use a **2-bit select signal** to choose the operation.

Steps

1. Write Verilog HDL code for the ALU.

2. Develop a testbench and simulate the design.
3. Perform synthesis **without any timing constraints**.
4. Generate a **timing report**.

Record the following

- Worst Negative Slack (WNS)
- Critical Path Delay
- LUT count
- Maximum Frequency

Problem 2: Applying Clock Constraint

Task

Apply a **clock constraint of 10 ns (100 MHz)**.

Example constraint:

```
create_clock -period 10 [get_ports clk]
```

Steps

1. Add the constraint in the **XDC file**.
2. Run synthesis and implementation again.
3. Generate a new timing report.

Observe

- Change in slack
- Change in critical path delay
- Change in LUT usage
- Maximum achievable frequency

Problem 3: Tight Timing Constraint

Task

Reduce the clock period to **5 ns (200 MHz)**.

```
create_clock -period 5 [get_ports clk]
```

Observe

- Whether timing requirements are met
- Increase in hardware resources
- Change in optimization strategy
- Updated slack value

Problem 4: Input and Output Delay Constraints

Apply the following constraints:

```
set_input_delay 2 -clock clk [all_inputs]
```

```
set_output_delay 2 -clock clk [all_outputs]
```

Record

- New slack value
- Updated critical path
- Impact on timing closure

Problem 5: Constraint-Based Optimization

Run optimization directives such as:

Area Optimization

Goal: Reduce hardware utilization.

Performance Optimization

Goal: Improve circuit speed.

Compare Results

Optimization Mode	LUT Count	Registers	Critical Path	Max Frequency	Slack
Default					
Performance					
Area					

Tools Required

- Vivado Design Suite
- Verilog HDL

Task Description

Students must:

1. Write Verilog modules.
2. Verify functionality through simulation.
3. Apply timing constraints using XDC.

4. Perform synthesis and implementation.
5. Generate timing and utilization reports.
6. Compare constrained and unconstrained designs.

Expected Outcome

- Understanding of **Static Timing Analysis**
- Ability to apply **clock and I/O constraints**
- Identification of **critical path and slack**
- Understanding the **trade-off between speed and area**

Evaluation Scheme (10 Marks)

Component	Marks
Verilog Coding	3
Functional Simulation	2
Timing Analysis	2
Optimization Comparison	2
Viva / Explanation	1